



Igniters Innovation Challenge 2026 | Engineering Track

An exclusive Engineering Track

Hosted by: Suzuki R&D Center

TEAM WOOKIES

Raunak Jain | Abhishek Aby | Divya

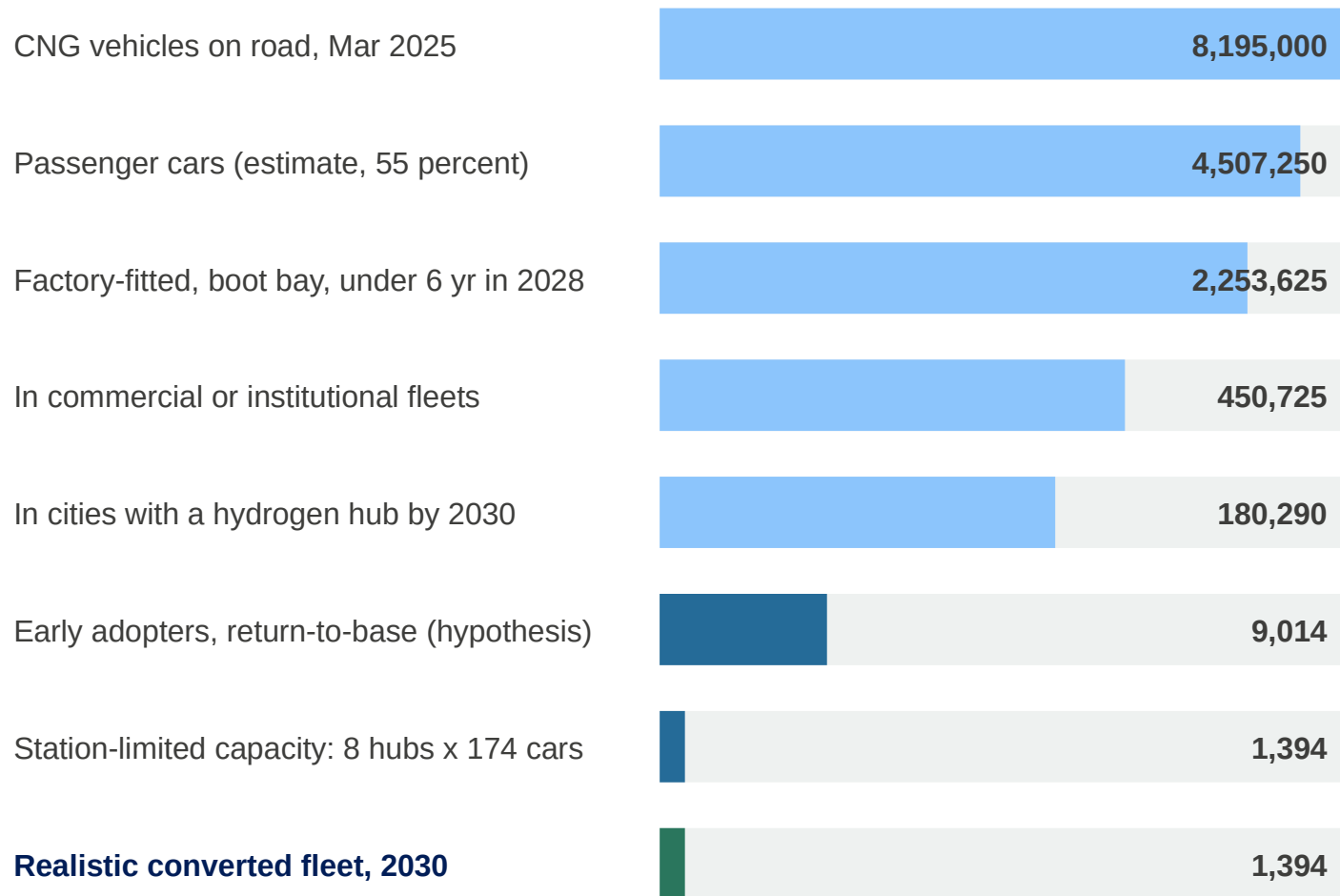
NIT Warangal

Recommendation: validate before selling. Run a 30-car hydrogen retrofit pilot in a captive Faridabad fleet and scale only if delivered hydrogen falls below about ₹190 per kg

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The brief asks whether retrofits can be a safe, affordable, viable complement to BEVs by 2030. Evidence answer: safe with engineering, not affordable at unsubsidised hydrogen prices, viable only as a station-limited niche.

Stations, not demand, set the niche: bottom-up funnel to 2030 (model)



Bar length on a log10 scale (1,000 to 8.2 million). One 500 kg/day hub at 70 percent utilisation serves 174 cars at 2.0 kg per car per day; demand (9,000) exceeds station capacity (1,394) sevenfold. Round 1 said 1,00,000 cars.

SAFE: yes, with engineering	AFFORDABLE: not by 2030	COMPLEMENT: niche only
Composite cylinder, thermal relief, twin sensors, lean burn with NOx control. No approval route exists for a hydrogen retrofit today: gate 1.	₹13.7/km converted vs ₹5.2/km on CNG at ₹800/kg (2028). CNG fuel parity needs ₹188/kg; cheapest 2030 forecast about ₹360/kg.	A compact BEV costs ₹5.6/km including purchase. Hydrogen retrofits fit captive fleets beside hydrogen hubs, about 1,400 cars by 2030.

Initial customer and use case

Donor	Maruti Suzuki Dzire Tour S CNG, 3 to 5 years old, factory bi-fuel; petrol stays as automatic backup
Fleet	Captive, return-to-base institutional and contract fleets: PSU and corporate staff transport, airport contracts
Duty	150 to 210 km per day, 330 days; two depot fills per day give about 120 km on hydrogen
Location	Faridabad, Delhi-NCR: IOCL R&D hydrogen dispensing point and the MNRE Sahibabad-Faridabad-Delhi corridor (sanctioned Mar 2025)
Excluded	Delhi app-cabs: aggregator scheme counts only EVs; CAQM allows only CNG or EV new aggregator cars from Jan 2026
Second hub	Ahmedabad-Vadodara-Surat corridor; Gujarat funds 30 percent of the first 20 hydrogen stations

Decision metrics (model, 2028 decision, 5 years)

₹13.7 vs 5.2 vs 5.6 per km: retrofit vs stay on CNG vs BEV (levelised, all-in)	₹188/kg delivered hydrogen price for fuel parity with CNG
61 km per hydrogen fill (1.12 kg usable); 555 km petrol backup	~1,400 cars conditional 2030 fleet, station-limited (Round 1: 1,00,000)

Conditions for success and where they stand today

Retrofit approval route defined (gate 1, Mar 2027)	Gap
None exists; AIS-195 covers new vehicles only	
Delivered hydrogen at or below ₹250/kg by 2030 with a path to ₹188	Not met
Model hub ₹584; CEEW central ₹360	
Certified kit at or below ₹2.5 lakh installed	Not met
₹3.2 lakh at 10,000/yr; ₹7.7 lakh pilot	
NOx at or below 60 mg/km on the BS-VI cycle	Target
Lean burn plus NOx trap provision; not yet tested	
Three fleets willing to accept 48 percent power loss and a smaller boot	Hypothesis
No interviews yet; Phase 0 buys 30	

What changed since Round 1

Fallback fuel	petrol already fitted, not a second CNG cylinder
Price basis	pump price (₹584 to 1,433/kg), not production cost (₹160 to 200)
Rollout	3 prototypes, 30 pilot cars, about 1,400 by 2030, not 1,00,000
Stop rules	four gates with proceed, redesign or stop criteria

Sources: PNGRB citing Vahan, 81.95 lakh CNG vehicles [W3-S2]; Delhi aggregator scheme [W3-S17]; CAQM direction [W3-S19]; IOCL Faridabad [W3-S46]; MNRE corridors, PIB 3 Mar 2025 [W3-S36]; Gujarat policy [W4-S76]; CEEW 2025 [W4-S22]; workbook sheets Market, TCO, Energy_Range.

Technology choice: H2-ICE is the only route that is a retrofit in cost and scope, but its fuel economics are the weakest; a fuel-cell conversion costs as much as a new BEV

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Matched criteria for the same donor and duty: Dzire Tour S CNG, 55,000 km per year, owner decision in 2028. Model outputs unless a source ID is given; estimates are labelled.

Criterion	H2-ICE retrofit (selected)	Fuel-cell conversion	New compact BEV, donor sold	Continue on CNG
Scope of change	Cylinder, lines, injectors, ECU, sensors; engine and petrol system retained	Whole powertrain: stack, battery, motor, inverter, cooling, tank, controls	New vehicle; donor sold for about ₹3.5 lakh	None
Cost to the owner, 2028	₹3.2 lakh installed at 10,000/yr (₹7.7 lakh pilot)	₹15 to 25 lakh pilot; ₹8 to 12 lakh at volume (estimate, W4)	₹10.5 lakh (Tiago EV to Punch EV class) [W3-S27]	₹0
Energy cost per km, 2028	₹12.0 blended; hydrogen km ₹14.8 at ₹800/kg	About half of H2-ICE per hydrogen km (stack efficiency)	₹1.5 at ₹6 depot, ₹20 public per kWh	₹3.6 at ₹87/kg
All-in cost per km, 5 years	₹13.7	Not modelled (capital above BEV)	₹5.6	₹5.2
Range per fill or charge	61 km on hydrogen plus 555 km petrol	About 2x H2-ICE on the same tank	187 to 260 km, real-world tests [W3-S30]	About 250 km at 25 km/kg
Refuel or recharge time	3 to 5 min, 350 bar non-cooled fill	3 to 5 min	About 1 h DC 10 to 80 percent; 6 to 8 h AC	3 to 5 min plus queue
Peak power vs petrol	About 52 percent, lean burn (city duty)	Restored by electric drive	Higher	About 85 percent
Tailpipe	Zero CO2 on hydrogen km; NOx must be controlled	Water vapour only	Zero	About 105 g CO2/km
Approval route today	None for retrofits; AIS-195 covers new H2-ICE vehicles	None; AIS-157 scope excludes retrofitted FCEVs	Established	Established
Supplier readiness	Injectors, valves, regulators production-intent; car-size cylinders scarce	Imported stacks; no light-duty supply chain in India	Mature	Mature

Global deployments: what is real, and the lesson	
Toyota Mirai (FCEV car) US sales 210 in 2025; fuel credits needed [W3-S51]	Commercial
Hyundai Nexo gen 2 (FCEV) Korea only at scale, subsidy-dependent [W3-S52]	Commercial
Toyota hydrogen HiAce (H2-ICE van) Depot fuelled; under 200 km; SCR for NOx [W3-S55]	Pilot
Suzuki-AVL hydrogen Swift (H2-ICE car) 1.4 L DI, 100 kW, Vienna May 2026; not a retrofit	Prototype
KEYOU truck conversions (H2-ICE) Only commercial retrofit model; Daimler 2027 [W3-S61]	Pilot
JCB hydrogen engine (off-highway) EU type approval May 2025 [W3-S60]	Approved
Stellantis hydrogen vans (FCEV) Cancelled Jul 2025: infrastructure, capital [W3-S64]	Withdrawn
Shell CA, H2 Mobility DE car stations 43 car stations closed 2024 to 2025 [W3-S65, S66]	Withdrawn
Status: Prototype = demonstrator; Pilot = fleet trial with data; Commercial = product on sale; Withdrawn = programme or stations closed.	

Why the selected route fits the donor, the customer and the question

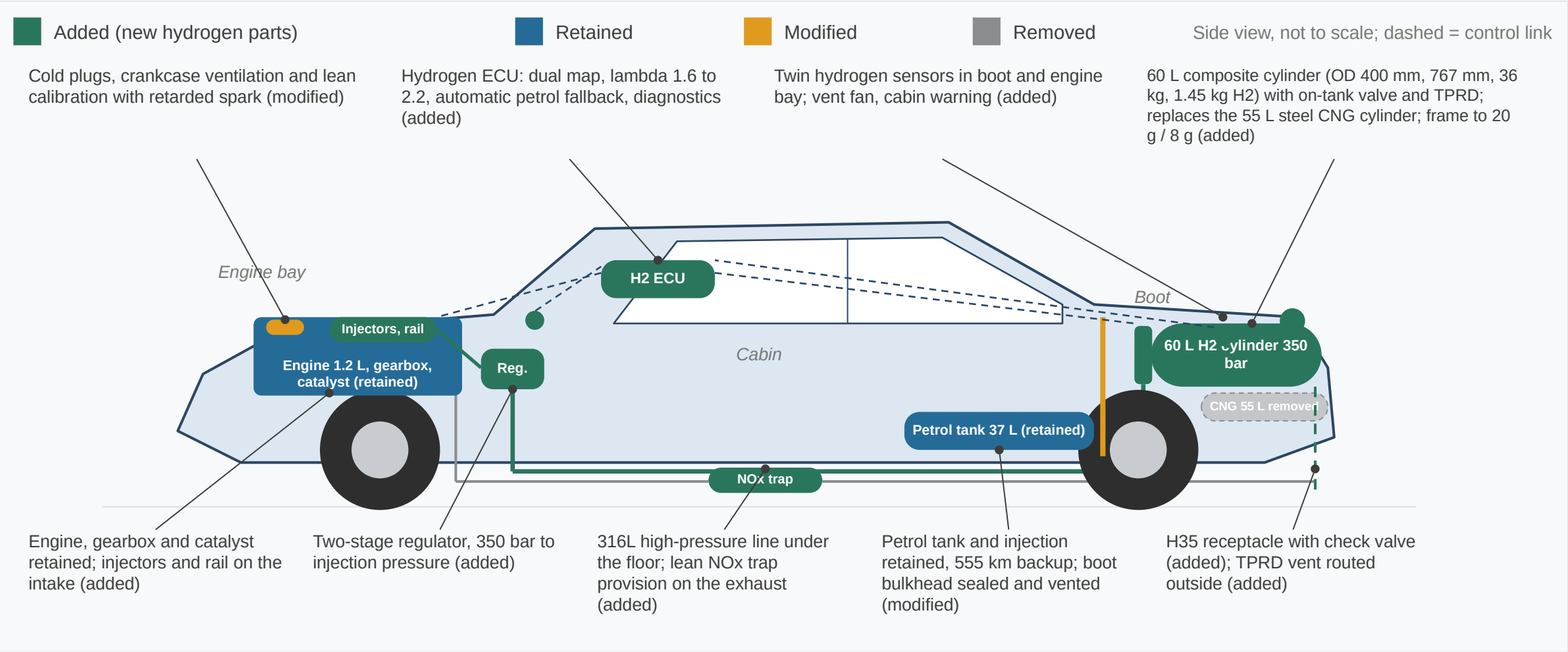
Donor fit 1.2 L four-cylinder SI engine with CNG-hardened seats, a 55 L cylinder bay and a 37 L petrol tank: only the fuel system is new; engine, gearbox, body and petrol backup are retained.	Customer fit Return-to-base duty with two depot fills makes a 61 km hydrogen range workable for 73 percent of a 167 km day; petrol covers the rest, so no car is stranded by a missing station.	Economic fit The only route whose cost is a kit (₹3.2 lakh) rather than a vehicle (₹8 to 25 lakh). Its case rests on one variable, the delivered hydrogen price, which the pilot is built to measure.
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Retrofit engineering: storage, fuel delivery, controls and safety are new; the engine, petrol system and body are retained. One fill gives about 61 km; power falls by about half

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Donor: Maruti Suzuki Dzire Tour S CNG, 1.2 L K-series, 60 kW on petrol, 55 L steel CNG cylinder in the boot, 37 L petrol tank, 163 mm ground clearance. Concept schematic, not validated CAD.



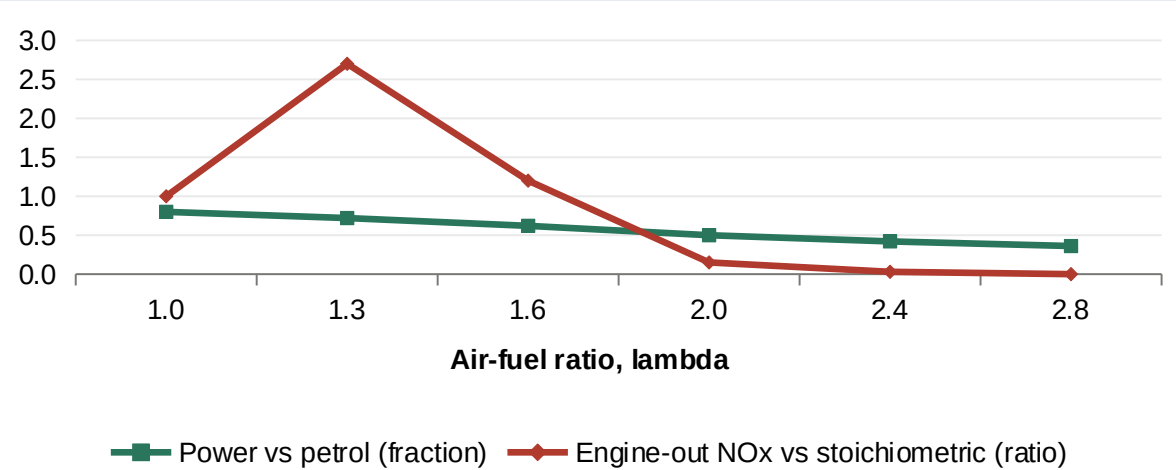
1.12 kg usable H2 per fill (1.44 kg nominal, 85 percent fill, 20 bar residual)	61 km per fill at 1.85 kg/100 km; 105 km at 700 bar	~121 km per day on hydrogen with two depot fills (73 percent of a 167 km day)	42 PS peak in hydrogen mode, 48 percent below petrol; petrol switch at high load	-13 kg net kerb mass; payload unchanged	~150 L boot left (about 200 L with CNG); seats unchanged	3 days conversion downtime per car (14 labour hours at volume)
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What cannot be reused, and the injection choice

Removed, not reusable: the 55 L steel CNG cylinder (ISO 11439 steels are limited to 2 percent hydrogen), CNG regulator, rail, injectors, hoses and lines. Hydrogen parts need UN R134 or ISO 19881 class qualification [W1-S11a].

Port injection timed after intake-valve closing prevents backfire and pre-ignition but costs power; direct injection (Suzuki-AVL Swift prototype) recovers it but is not a retrofit. Durability to prove: valve seats and guides, lubricant ash, sensor drift.

The central trade-off: lean burn buys low NOx at the cost of power



Schematic curves through literature anchors, not measured data: port injection loses about 20 percent power at lambda 1 and a further 35 to 40 percent at lambda 2 [W1-S9a, S9b]; NOx peaks near lambda 1.3 at about 2.7 times the stoichiometric value and falls to near zero above lambda 2 to 2.7 [W1-S9h, S9b]. Accelerations leave the lean window, hence the NOx trap provision [W4-S70].

Packaging consequences (concept, mock-up in Phase 0)

Item	CNG today	Hydrogen	Effect
Cylinder OD x length	316 x ~900 mm	400 x 767 mm	Boot loses ~50 L more
Cylinder mass	~55 kg steel	36 kg composite	Kerb mass -13 kg
Fuel on board	~9 kg CNG	1.12 kg usable H2	35 percent of energy
Seats / luggage	5 / ~200 L	5 / ~150 L	Two bags become one

Consumption cross-check (kg H2 per 100 km)

Vehicle	kg/100 km	Basis
This retrofit, fleet use (model)	1.85	0.87 x CNG efficiency at 25 km/kg
Ford Focus H2RV (2.3 L, hybrid)	1.4	claimed, 2003 [W1-S12c]
Mazda RX-8 Hydrogen RE, rotary	2.4	claimed, 2006 [W1-S12b]
Toyota hydrogen HiAce (3.4 L van)	3.1	6.2 kg for 200 km [W1-S12e]

Safety and approval: each hazard has a known prevention, detection and test; the unresolved item is approval, because no route exists yet for a hydrogen retrofit of a car

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Hydrogen: flammable from 4 to 75 percent in air at 0.017 mJ, diffuses 3.8 times faster than natural gas, burns almost invisibly [W2-S43, S44]. Different risks, not automatically higher, if designed for.

Hazard	Prevention (design)	Detection and response	Evidence required	Evidence type
Leak into boot, cabin, garage	Sealed bulkhead, vented boot, no ignition sources in the bay, valve closed with ignition off	Two sensors (1 and 2 percent), tank valve closes within 1 s, switch to petrol, cabin warning	Below 4 percent in cabin, below 8 percent at vents (GTR 13 / AIS-195); parked enclosed-space test	Type test
Embrittlement, permeation, ageing	316L lines, hydrogen elastomers, composite liner; no CNG steel parts reused	Leak check at service; 3-yearly requalification; 15-year life	ISO 19881 / UN R134 supplier qualification; cycling and permeation	Standard
Crash damage	Cylinder inside the crush-free zone; frame to 20 g / 8 g	Inertia switch closes the tank valve; post-crash leak limit	Sled test; full-vehicle rear crash under AIS-195	Type test
Overpressure and fire	TPRD vents away from occupants and wheel wells; burst ratio 2.25 x NWP	TPRD opens on heat: controlled release, not rupture	Bonfire, burst, drop, cycling (GTR 13); vent-routing review	Type test
Detection or control failure	Redundant sensors, ECU watchdog, manual shut-off, limp-home on petrol	Fault codes, forced petrol mode, service lock-out	Fault injection and hardware-in-the-loop; 500 h engine durability	Programme test
Workshop and refuelling	NFPA 2 ventilation (1 cfm per sq ft), detection, permit-to-work, PESO-trained staff	Dispenser and bay interlocks; nitrogen purge for venting	PESO Form-H licence; SAE J2601 fill protocol; incident drills	Licence
Combustion emissions (NOx)	Lean burn lambda 1.6 to 2.2, retarded spark, NOx trap provision, load limiting	Lambda sensor and OBD	BS-VI 60 mg/km on the certification cycle; quarterly pilot tests	Target

NOx is the emission that matters: hydrogen combustion still forms thermal NOx (US DOE [W2-S52]). Lean burn above lambda 2 is near zero at steady state, but accelerations leave that window, so the kit carries a lean NOx trap provision and gate 2 requires 60 mg/km or better on the BS-VI cycle. Target, not a result. No 'approved' or 'fully safe' claim is made; safety evidence comes from type tests and standards, not from incident-free pilot operation.

Approval sequence: what exists and what is missing

1

New-vehicle hydrogen ICE standard

AIS-195:2023 (Amd 1, May 2025) via CMVR Rule 125M, GSR 746(E) [W2-S13]

Established

2

Retrofit gateway in law

MV Act s.52; Rule 126 proviso; Rule 115 kit machinery written for CNG and LPG [W2-S3, S9]

Established

3

Hydrogen retrofit kit standard and emission procedure

None; AIS-157 excludes retrofitted FCEVs; AIS-195 retrofit position unverifiable [W2-S11]

Gap

4

Car-size cylinder approval (PESO)

Gas Cylinder Rules 2025 cover on-board hydrogen; only 150 to 250 L truck cylinders approved [W2-S24, S30]

Unresolved

5

Dispensing licence

PESO Form-H online since May 2025; SMPV(U) 2025 for bulk storage [W2-S27]

Established

6

Registration, insurance, warranty

Form 22D endorsement needs an approved kit; IMT-25 covers CNG/LPG only; OEM warranty void [W2-S53, S54]

Unresolved

7

Trial route for the pilot

Trade certificate plus test-agency evaluation (Mirai 2022, Tata trucks 2025); MNRE scheme names 4-wheelers

Established

Gate 1 (March 2027): a written regulator and test-agency view on the retrofit route, or the programme stops. Legend: green = in force; amber = exists but not for this case; red = missing instrument.

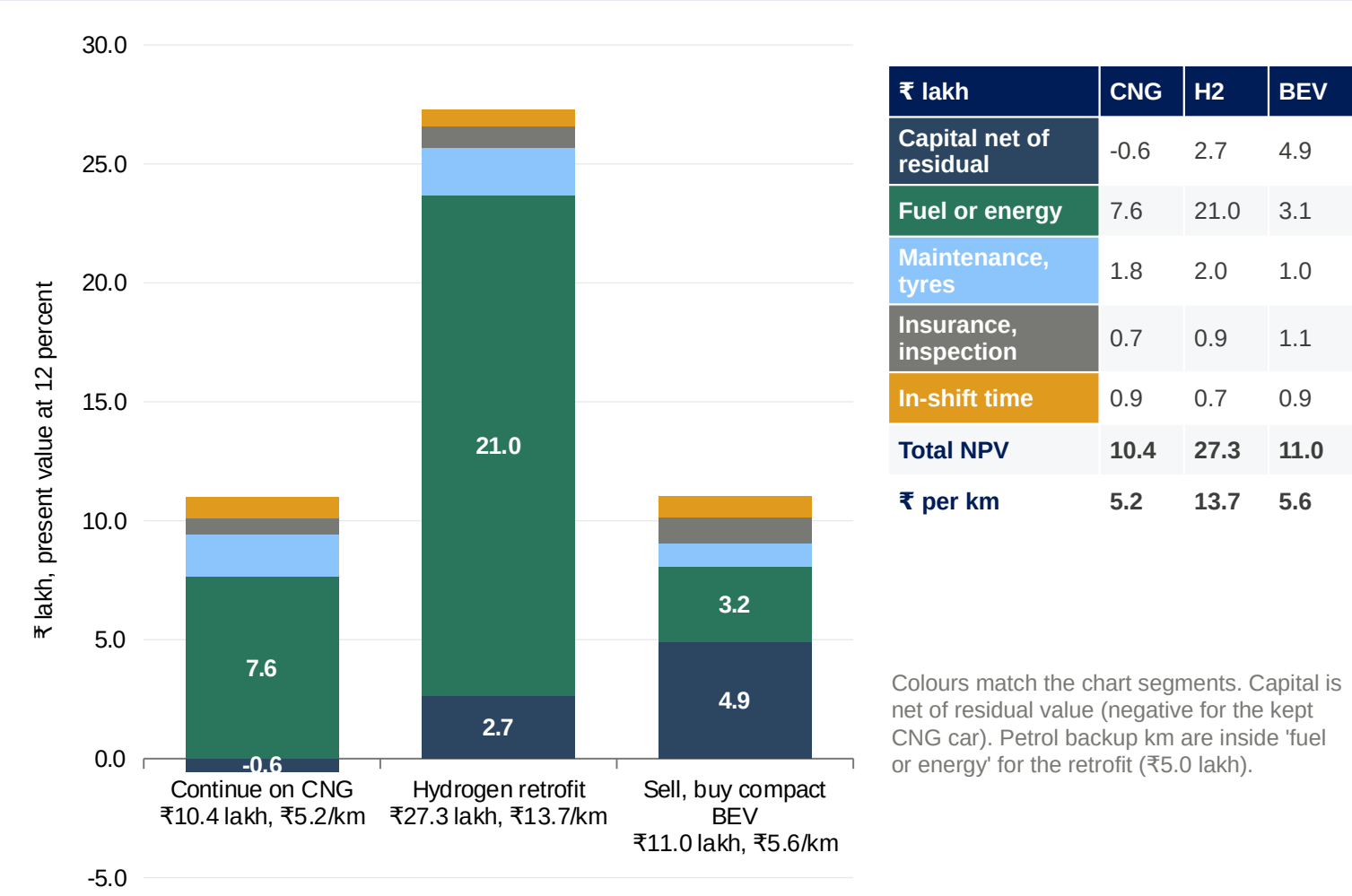
Economics: over five years the converted car costs the owner ₹16.8 lakh more than staying on CNG and ₹16.2 lakh more than switching to a BEV; fuel is the whole story

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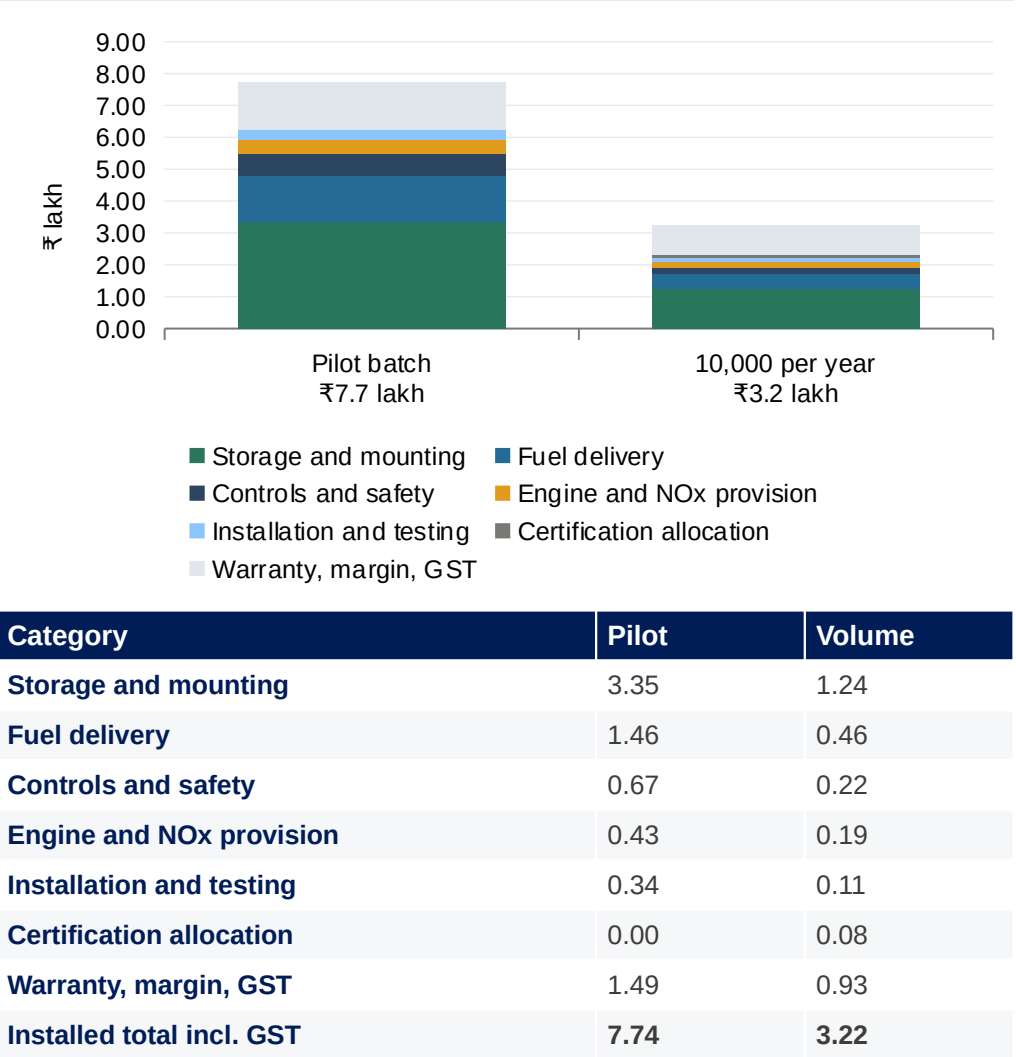
Fair comparison for an owner deciding in 2028 with a 3-year-old Dzire Tour S CNG. Same horizon, mileage, discount rate and time costs; donor sale credited to the BEV option; residual values credited to all.

Five-year cost of ownership by category, ₹ lakh NPV (model, base case)



Fuel or energy: hydrogen km ₹14.8/km at ₹800/kg falling 10 percent a year, petrol backup km ₹6.8, CNG ₹3.5, BEV ₹1.5. Negative capital for the CNG car is its residual value.

Conversion cost by category, ₹ lakh installed (BOM sheet)



Extra cost vs staying on CNG, ₹ lakh over 5 years

₹/kg \ km/yr	30,000	50,000	70,000
₹150	+3.6	+3.8	+4.0
₹250	+4.7	+5.6	+6.5
₹400	+6.3	+8.3	+10.3
₹600	+8.5	+11.9	+15.4
₹800	+10.7	+15.6	+20.5

Every cell is positive: no cell turns negative in the workbook grid down to ₹100/kg. More mileage widens the gap while hydrogen is above ₹188/kg.

Break-even indicators (model)

CNG fuel parity	₹188/kg delivered
Petrol fuel parity	₹368/kg
Full-TCO break-even vs CNG	none in 5 years at any price
Full-TCO break-even vs BEV	none at base; needs about ₹150/kg, ₹2 lakh kit, 85 percent hydrogen km
Abatement cost vs CNG	₹129,000/t CO2e at base; about ₹45,000/t at ₹250/kg (credits ₹1,350 to 2,800/t)

Assumptions used in every exhibit on this slide (Inputs sheet; downside and upside cases in the annexure)

Decision 2028, 5 years

55,110 km/yr

Discount 12 percent

H2 ₹800/kg, minus 10 percent/yr

CNG ₹87/kg at 25 km/kg

Petrol ₹102/L

H2 1.85 kg/100 km, 65 percent of km

Power ₹6 depot, ₹20 public per kWh

BEV ₹10.5 lakh, resale ₹4.2 lakh

Donor sale ₹3.5 lakh

Time ₹150/h in shift

Reading: the retrofit is a decarbonisation cost, not a saving. Hydrogen km cost ₹14.8 against ₹3.5 on CNG, and the 35 percent of km on petrol backup add ₹5.0 lakh. It needs a paying counterparty or a cheaper molecule than any 2030 forecast found (CEEW ₹360/kg delivered; refinery-gate record ₹279/kg before compression).

Sources: fuel prices Sep 2026 [W4-S16, S19]; DERC tariff, public charging [W4-S20, S21]; BEV prices, tests, warranties [W3-S27, S30; W4-S26]; NITI Aayog mileage [W4-S30]; CEEW 2025 [W4-S22]; Numaligarh tender [W3-S39]; carbon credits [W4-S73]; TCO, BOM and Sensitivities sheets.

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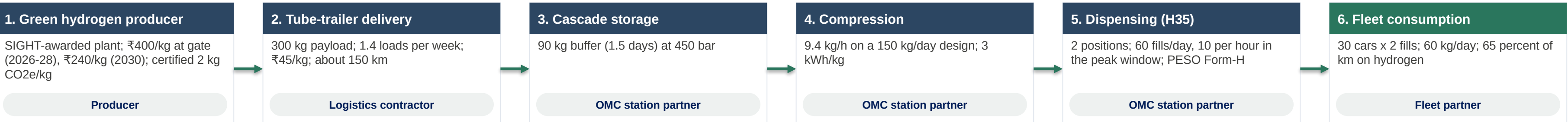
Hydrogen supply: a 30-car pilot needs about 60 kg a day; delivered hydrogen costs about ₹1,433/kg at a pilot point and ₹584/kg at a 2030 hub, and utilisation alone cannot reach parity

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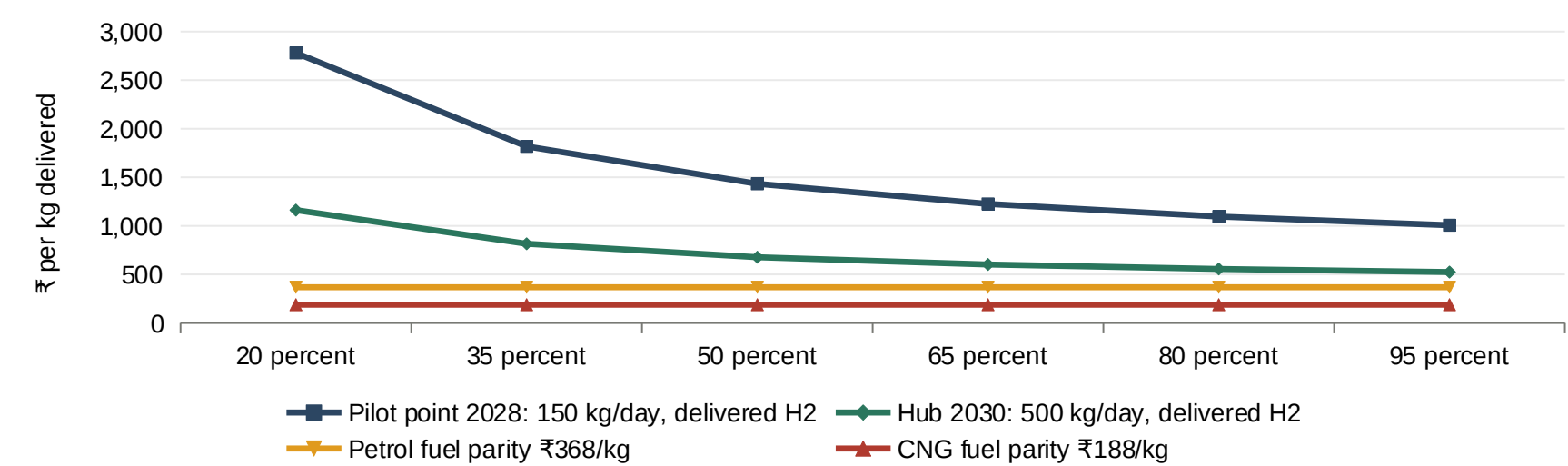
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Pilot setting: Faridabad, Delhi-NCR (IOCL R&D dispensing since the 2022 Mirai pilot; MNRE corridor sanctioned March 2025, 18 to 24 months to commission). No station is public or confirmed available: access is a proposed partnership.

Supply-to-fleet flow with pilot quantities (model, H2_Supply sheet) and proposed owners



Delivered price falls with utilisation but never reaches CNG parity (model)



At 95 percent utilisation the hub price is still ₹524/kg: production, transport and compression alone (₹312/kg) exceed CNG parity. Station utilisation is the axis; capex ₹10 crore (pilot) and ₹12 crore (hub) recovered over 10 years at 12 percent.

Delivered price build-up, ₹ per kg (model): pilot point 2028 versus 2030 hub

Component	Pilot point, 150 kg/day at 50 percent	Hub 2030, 500 kg/day at 70 percent
Production at plant gate (green)	400	240
Tube-trailer transport	45	45
Compression electricity	27	27
Station capital recovery (10 years, 12 percent)	647	166
Station fixed O&M	146	38
Losses, margin and taxes	168	69
Delivered price at the dispenser	1,433	584

Ownership and risk (proposed, not committed)

Function	Proposed holder	Risk carried
Vehicles	Fleet partner (PSU, corporate or airport contract)	Residual value, backed by programme buy-back
Kits	SRDI-led consortium; kit integrator warrants	Development and warranty
Dispensing point	OMC partner (proposed IOCL, Faridabad), Form-H licence	Utilisation, covered by take-or-pay from the fleet
Fuel	SIGHT-awarded producer by tube trailer	Supply reliability; buffer covers 1.5 days
Service	Two hydrogen-ready workshops; PESO test station	Safety compliance and training
Evaluation	IIT Delhi or CEEW (proposed)	Credibility of results

Main bottleneck: not station throughput (2 dispensers cover 60 fills a day) but the hydrogen price stack. Delivered gas keeps pilot capital to a dispensing point; on-site electrolysis is justified only at hub scale. Petrol backup covers supply interruptions.

Station sizing from the fleet (H2_Supply sheet)

- 30 cars x 167 km x 0.0185 kg/km x 0.65 = 60 kg/day
- 60 fills/day, 50 percent in a 3 h window = 10/h, 2 dispensers
- Compressor 9.4 kg/h over 16 h; buffer 90 kg (1.5 days)
- Cars alone use 40 percent of capacity; 50 percent assumed with bus and truck users
- One 2030 hub serves 174 cars at 70 percent utilisation

Roadmap to 2030: four gated phases across five workstreams, ₹37.6 crore including contingency, with a proceed, redesign or stop decision at each gate

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October 2026 to September 2030. Suzuki R&D Center India proposed as engineering lead; public support sought under the MNRE transport pilot scheme, which names buses, trucks and 4-wheelers (PIB 14 Feb 2024).

	2026	2027			2028			2029			2030		
Decision gates		G1 Mar 2027			G2 Mar 2028			G3 Mar 2029			G4 Sep 2030		
Phase and budget (₹ crore)	Phase 0: feasibility ₹0.9 cr	Phase 1: engineering and controlled validation ₹8.5 cr			Phase 2: certification and pilot build ₹15.2 cr			Phase 3: pilot operation ₹8.1 cr			Cond.		
Engineering Owner: SRDI lead	Packaging mock-up, CAD, hazard analysis	500 h test-cell programme; 3 prototype conversions			30 conversions; installer training			Field support, calibration updates					
Testing and approval Owner: ARAI or ICAT; PESO	Route consultation: AISC, MoRTH, ARAI, ICAT, PESO	Storage qualification; leak, ventilation, crash-structure tests			Kit type approval; 80,000 km durability on 2 cars			Quarterly emission and safety tests					
Fuel supply Owner: OMC partner	Station partner letter; supply options	Supply contract; Form-H application			Dispensing point commissioned			Deliveries 1.4 loads/week; utilisation log					
Fleet operations Owner: Fleet partner	30 interviews; telematics on 20 cars	Driver and workshop training plan			2 workshops upgraded; 30 cars inducted			18 months with 10 CNG and 10 BEV comparison cars					
Commercial validation Owner: SRDI with evaluator	Demand evidence from 3 fleets	Cost model with supplier quotations			MNRE sanction; insurer endorsement			Independent evaluation; scale decision					

Dependencies: gate 1 needs the regulator view and a station partner letter; Phase 1 needs car-size cylinder samples (Luxfer, Quantum, NPROXX, Time Technoplast) and hydrogen injectors; Phase 2 needs the MNRE sanction, PESO approvals and corridor station timing; Phase 3 needs the fuel contract and an insurer endorsement. Phase 4 (from Oct 2030, grey) is conditional: 1,000 to 1,500 cars at up to 8 hubs.

Gate	Proceed if	Otherwise
G1 Mar 2027	Written route from regulator and test agency; station partner secured; demand evidence from 3 fleets	Stop
G2 Mar 2028	At or below 2.2 kg/100 km; NOx at or below 60 mg/km on cycle; no abnormal combustion in 500 h; safety tests passed	Redesign once, then stop
G3 Mar 2029	Kit approval granted; dispensing point commissioned within 6 months of plan	Pause
G4 Sep 2030	Availability 95 percent; zero reportable hydrogen incidents; measured ₹/km published; delivered hydrogen at or below ₹250/kg with a credible path to ₹190/kg	Stop and publish

Proposed funding of ₹37.6 crore (shares are proposals)

40%	35%	15%	10%
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SRDI and Maruti Suzuki ₹15.0 cr | Public: MNRE pilot or R&D scheme, state policy ₹13.1 cr | OMC partner ₹5.6 cr | Fleet partner ₹3.8 cr. Why Suzuki's role is plausible: CNG engine and calibration know-how, 70 percent of India's CNG cars, its own hydrogen combustion prototypes (Swift with AVL, Burgman). No endorsement is implied.

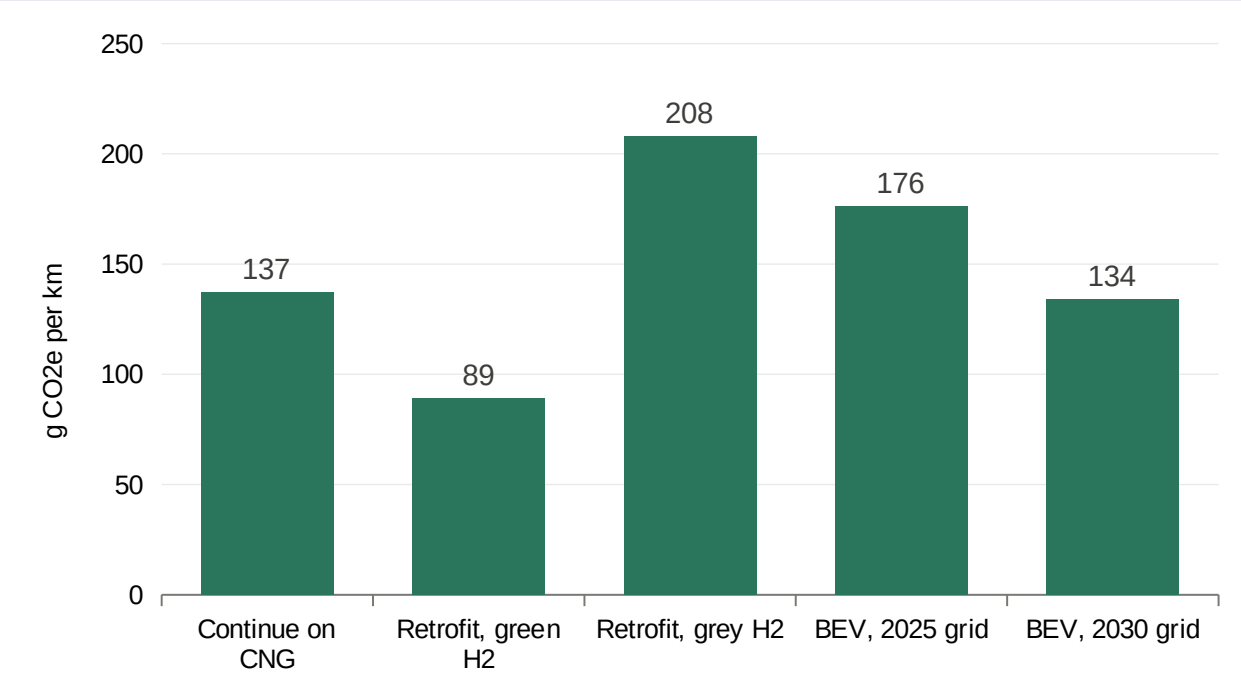
Impact is real but small by 2030: about 1,400 cars and 3.6 kt CO2e a year, conditional on the gates. The decision requested is ₹0.9 crore for Phase 0

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Matched service (same car, same duty) for every comparison; projections are labelled as modelled, pilot target or conditional scenario, never as achieved results.

Emissions per km, well-to-wheel with manufacturing (model)



Green-hydrogen retrofit is 35 percent below CNG with petrol backup km included; grid-electrolysis hydrogen would be 533 g/km, four times CNG. Reversal: hydrogen above 5.9 kg CO2e/kg makes the retrofit worse than CNG; grid below 0.29 kg/kWh at the meter makes the BEV cleaner (CEA path about 0.5 by 2032). NOx remains.

Scenario table (per car unless stated)

Metric	Baseline: keep CNG	Modelled retrofit, base	Pilot target 2029-30	Conditional 2030 (about 1,400 cars)
CO2e, g/km	137	89	at or below 95	3,645 t/yr avoided
Hydrogen use	0	1.85 kg/100 km	at or below 2.2	924 t/yr demand
Cost per km, all-in	₹5.2	₹13.7	measured and published	depends on hydrogen price gate
Availability	reference	not yet measured	at or above 95 percent	at or above 95 percent

Pilot scorecard: metric, success threshold, measurement

Metric	Threshold (gate 4)	How measured
Consumption	at or below 2.2 kg/100 km in fleet use	Dispenser mass logs, CAN flow
Range per fill	at or above 55 km	Trip logs, GPS
Hydrogen share of km	at or above 60 percent	Telematics, fuel mode log
Availability	at or above 95 percent	Downtime by cause
Safety	zero reportable incidents; all alarms closed	Incident register, sensor logs
NOx	at or below 60 mg/km on cycle	Test agency, quarterly
Maintenance cost	within ₹1.2/km	Job cards
Refuelling time	under 6 min per fill	Dispenser logs
Delivered hydrogen	at or below ₹250/kg, path to ₹190	Fuel invoices, station utilisation
Acceptance	fleet managers and drivers accept power and boot	Quarterly interviews, willingness to pay

Major unresolved risks (ranked)

Risk	Why it matters	Mitigation
Delivered hydrogen price	₹360 to ₹1,430/kg across sources; decides the scale case	Price gate; take-or-pay; pilot fuel as programme cost
Approval route	No instrument for a hydrogen retrofit kit	Gate 1 consultation; trial under trade certificate
NOx on transients	Lean window lost under acceleration	NOx trap provision; load limiting; cycle test at gate 2
Cylinder supply	No car-size PESO-approved cylinder	Samples from 4 suppliers in Phase 0; PESO prototype approval
Customer demand	Hypothesis; power and boot loss may be refused	30 interviews; gate 1 needs 3 fleets

Decision requested

- 1

Approve Phase 0: ₹0.9 crore, six months
regulator consultations, 30 fleet interviews, telematics on 20 cars, boot mock-up; no commitment beyond gate 1
- 2

Endorse a scheme application
MNRE transport pilot scheme, 4-wheeler category, with ARAI as implementing agency; Gujarat station subsidy for the second hub
- 3

Nominate roles
SRDI engineering lead; fleet and fuel partner shortlist (proposed IOCL Faridabad; PSU and airport-contract fleets)
- 4

Start HCNG in parallel
where a CGD company blends to IS 17314: legal since 2020, no vehicle change, about 6 percent of energy from hydrogen; an air-quality step

What would make us stop

- No written regulator route by March 2027
- Backfire or NOx not controlled within 500 test-cell hours
- Delivered hydrogen above ₹250/kg in 2030 with no path to ₹190/kg
- Fleets refuse the power and boot loss in Phase 0 interviews

H2-SHIFT keeps India's hydrogen retrofit option open at low cost, states the price at which it works, and hands the 2030 scale decision to evidence.

Key sources by theme (IDs as used on the slides) and annexure index

Research cut-off 14 September 2026. The full register (309 entries with URLs, dates, sections and limitations) is in Annexure A2 and the workbook Sources sheet.

Policy, standards and approval

- W2-S13 MoRTH GSR 746(E), 16 Oct 2023: Rule 125M, AIS 195:2023 (Amd 1, May 2025)
- W2-S11 MoRTH GSR 579(E), 23 Sep 2020: AIS 157:2020; scope excludes retrofitted FCEVs
- W2-S1 MoRTH GSR 585(E), 25 Sep 2020: H-CNG per IS 17314:2019
- W2-S3, S9 CMVR Rules 115 and 126 (CNG/LPG kit provisions, retrofit proviso)
- W2-S24, S27 Gas Cylinders (Amendment) Rules 2025, GSR 225(E); PESO Form-H licensing (May 2025)
- W2-S42 to S45 UN GTR 13 Amend. 1 (2023); H2Tools and DOE properties; NFPA 2 (2023)

Hydrogen supply and cost

- PIB 2107795 (3 Mar 2025): 5 pilots, 37 vehicles, 9 stations, 10 routes, ₹208 crore
- PIB 2006052 (14 Feb 2024): scheme guidelines, buses, trucks and 4-wheelers, ₹496 crore; W3-S37 PIB 29 Jul 2026: 12 projects, 70 vehicles, 16 stations
- W4-S22 CEEW 2025: delivered green hydrogen USD 4.2/kg in 2030; above USD 10/kg today
- W4-S33, S38, S57 CEEW, SIGHT ammonia auctions, IEA GHR: production cost evidence
- W3-S39 Numaligarh refinery tender ₹279/kg; W4-S46 ICCT station cost by utilisation
- W3-S46, W4-S81 IOCL R&D Faridabad hydrogen dispensing

Market, fleet and customer

- W3-S2 PNGRB citing Vahan: 81.95 lakh CNG vehicles (Mar 2025)
- W3-S4, S6 MoPNG: 8,150 plus CNG stations; 17,700 to 18,000 by 2030 (targets)
- W3-S10, S12 Maruti Suzuki CNG sales: 6.12 to 6.2 lakh FY25, above 7 lakh FY26; 71.6 percent segment share
- W3-S17, S19 Delhi aggregator scheme 2023; CAQM NCR direction (Jan 2026)
- W4-S30 NITI Aayog (Feb 2026): 4W taxi mileage 70,000 km/yr
- W3-S27, S30 Tata Tiago EV, Punch EV, Xpres-T prices; Autocar India real-world tests

Emissions

- W4-S59, S60 CEA CO2 Baseline Database v21 (0.710 t/MWh, FY2024-25); National Electricity Plan 2022-32
- W4-S62 IPCC 2006 fuel factors (petrol, natural gas)
- W4-S63 ICCT 2025 lifecycle factors; battery about 70 kg CO2e/kWh
- W4-S64, S65 IEA hydrogen carbon intensities; MNRE Green Hydrogen Standard (2 kg CO2e/kg)
- W2-S52 US DOE: hydrogen combustion and NOx

Vehicle, components and combustion

- W1-S3a to S3c Maruti Suzuki Tour S and Dzire specifications and releases
- W1-S2 Luxfer G-Stor Pro H2 table (V060H35: 60 L, 350 bar, 36 kg, 1.45 kg)
- W1-S1 Hexagon Purus Type 4 cylinder table (Rev 08/22)
- W1-S9a, S9b, S9h PFI air displacement and power loss; lambda-NOx behaviour (IJHE 2024-25; Onorati et al.)
- W1-S11a ISO 11439 hydrogen limit for steel CNG cylinders
- W4-S1, S2 US DOE records 19008 and 24006: Type IV storage system cost

Global programmes

- W3-S51, S52 Toyota Mirai sales; Hyundai Nexo
- W3-S55, S56 Toyota hydrogen HiAce pilot; GR Corolla H2
- Fuel Cells Works, RushLane (May 2026): Suzuki-AVL hydrogen Swift prototype; W3-S58 Suzuki Burgman
- W3-S59 to S63 Cummins B6.7H India; JCB type approval; KEYOU; MAN; ULEMCo
- W3-S64 to S67 Stellantis (Jul 2025); Shell California (Feb 2024); H2 Mobility Germany (2025); GM-Honda

Annexure index (separate PDF and workbook)

Section	Content
A1	Claim audit: Round 1 versus evidence, with corrections
A2	Assumptions, evidence classes, full source register
A3	Cost model: BOM, TCO cash flows, break-even, sensitivities
A4	Packaging, energy balance and performance calculations
A5	Regulatory compliance matrix and hazard analysis
A6	Market sizing method and funnel
A7	Global benchmark evidence
A8	Pilot design, instrumentation, first 90 days
A9	Phase budgets, responsibilities, funding
A10	Judge Q&A, validation log, change log
Workbook	H2SHIFT model: 11 formula-driven sheets, recalculated

Model inputs by evidence class (98 inputs)

A 31	B 13	C 52	D 2
A 31 B 13 C 52 D 2 (A verified, B model, C assumption, D target)			

Evidence classes: A verified external evidence; B model estimate; C explicit assumption or range; D proposed target. Many originals were inaccessible through the build environment's network policy; entries marked 'excerpt' or 'summary' rest on indexed page text and should be re-verified before external use. No customer interviews, letters of interest, test results or partnerships are claimed.